

Progress Report 5

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Project Name: F2025_4495_071_CKa004 (Weather Driver)

Work Log Table

Date	Number of hours	Description of work done
Nov 9, 2025	5	Implemented add stops feature with route recalculation. Added stops state management in ActiveNavigationScreen. Fixed search, location picker, and map picker flows in AddStopsScreen. Route now recalculates through all stops using waypoints. Added numbered stop markers on map. Display stops in collapsible turn-by-turn instructions with highlights. Show origin address at top of navigation steps. Added remove button for each stop in header card. GitHub Commit: 718e49c: "Implement add stops feature with route recalculation"
Nov 10, 2025	4	Designed route selection UI/UX with horizontal scrollable cards. Planned implementation of route switching via map polylines. Researched Google Maps API for multiple route response handling. Created wireframes for route card component layout.
Nov 11, 2025	5	Implemented route selection functionality in RoutePreviewScreen. Added horizontal scrollable cards showing all available routes. Enabled tap-to-switch for route cards and map polylines. Added visual highlighting with blue border for selected route. Implemented map animation to fit selected route. Added "Fastest" badge for quickest route option. Merged praveshe-updates into main branch. GitHub Commits: c93a4c6: "Add route selection functionality in preview screen", ea191a2: "Merge praveshe-updates into main"
Nov 12, 2025	5.5	Implemented saved trips feature with persistent storage using AsyncStorage. Built new location picker screen with search, current location, and map options. Updated saved trips to show real-time distance, duration, and traffic info. Added ability to set and change saved locations (home, work, gym, school). Integrated with route preview for quick navigation. Replaced dummy suggestions with actual nearby places from Google API. GitHub Commits: 9c7044f: "Add saved trips feature with persistent storage", 19d242d: "Sync praveshe-updates with main", b63571e: "Comment out EditProfileScreen temporarily"
Nov 13, 2025	3	Code review and merge conflict resolution with team members. Reviewed pull requests from Chetan's branch. Assisted with merging changes to main branch. Tested integrated features for compatibility issues.

Date	Number of hours	Description of work done
Nov 14, 2025	4	Designed saved locations integration across all location picker screens. Planned useSavedLocations custom hook architecture. Analyzed SearchScreen, ChooseStartLocationScreen, ChooseDestinationScreen, and AddStopsScreen for integration points. Documented API data flow for saved locations.
Nov 15, 2025	6	Integrated saved locations across all location picker screens. Created useSavedLocations hook to fetch saved trips from AsyncStorage. Updated SearchScreen to display actual saved Home/Work/School/Gym locations. Updated ChooseStartLocationScreen, ChooseDestinationScreen, and AddStopsScreen to use real saved locations with coordinates. Added loading states and placeholder styling for unset locations. Implemented auto-refresh when screens come into focus. Implemented profile image picker with gallery and camera options. Added camera icon overlay on profile avatar. Installed expo-image-picker and react-native-country-picker-modal. GitHub Commits: a2225fd: "Integrate saved locations across location picker screens", 695ca67: "Add profile image picker and enhance edit profile functionality", 5e43f16: "Midterm Video Link File Uploaded"
Nov 16, 2025	3	Tested the saved locations feature end-to-end. Verified coordinates pass correctly for accurate routing. Fixed edge cases for unset location placeholders. Conducted cross-device testing for profile image picker functionality.
Nov 17, 2025	2.5	Bug fixes and performance optimization for the saved trips feature. Improved loading states for better user experience. Enhanced error handling for AsyncStorage operations. Cleaned up console logging from the development phase.
Nov 18, 2025	3	Supported team with EAS Build integration. Assisted with folder renaming for EAS compatibility. Reviewed authorizer and permission implementations. Tested build migration on the server for Google sign-in.
Nov 19, 2025	2	Reviewed the team's Google sign-in server build implementation. Tested authentication flow integration points. Documented remaining integration requirements.
Nov 20, 2025	3.5	Reviewed and tested login/signup screen implementations. Verified Google authentication button UI. Tested manual login and signup credential validation. Ensured consistent user experience across authentication methods.
Nov 21, 2025	2.5	Final testing of all implemented features. Verified saved trips persistence across app sessions. Confirmed route selection and profile features are working correctly. Prepared documentation for sprint review.
Nov 22, 2025	3	Reviewed all commits for the reporting period. Ensured all changes were properly merged to the praveshteam/updates branch—prepared progress report for submission.

Description of Work Done

During this reporting period (November 9-22, 2025), significant progress was made on the Weather Driver mobile navigation application, focusing on route selection functionality, saved trips feature implementation, profile enhancements, and cross-screen integration of saved locations.

Week 1 (November 9-15): The primary focus was on implementing three major features. First, the add stops feature was implemented, allowing users to add multiple waypoints to their route. This includes stop state management in `ActiveNavigationScreen`, route recalculation through waypoints using Google Maps API, numbered stop markers on the map, stops displayed in collapsible turn-by-turn instructions with visual highlights, and a remove button for each stop. Second, the route selection functionality was added to the `RoutePreviewScreen`, enabling users to choose between multiple route options provided by Google Maps. This feature includes horizontal scrollable route cards, tap-to-switch functionality on both cards and map polylines, visual highlighting for the selected route, automatic map animation when switching routes, and a “Fastest” badge indicator. Third, a comprehensive saved trips feature was implemented using `AsyncStorage` for persistent storage, allowing users to save frequently visited locations (home, work, gym, school) and quickly navigate to them. The feature includes real-time distance, duration, and traffic information display, and integrates with the route preview system. Additionally, profile enhancement work included implementing a profile image picker with both gallery and camera options, adding a camera icon overlay for intuitive photo selection, and installing necessary dependencies (`expo-image-picker`, `react-native-country-picker-modal`).

Week 2 (November 16-22): The focus shifted to integrating saved locations across all location picker screens in the application. A custom `useSavedLocations` hook was created to fetch and manage saved trips from `AsyncStorage`. The `SearchScreen`, `ChooseStartLocationScreen`, `ChooseDestinationScreen`, and `AddStopsScreen` were all updated to display and utilize actual saved location data with coordinates for accurate routing. Loading states and placeholder styling were added for unset locations, and auto-refresh functionality ensures data stays current when screens come into focus. This week also involved extensive testing, bug fixes, and supporting team members with EAS Build integration and Google authentication implementation.

Technical Skills Applied: React Native development with custom hooks, state management, and navigation - AsyncStorage implementation for persistent data storage - Google Maps API integration for multiple routes and real-time traffic data - Expo packages integration (expo-image-picker) for native device functionality - UI/UX design with scrollable components and visual feedback systems - Git version control with branch management and merge conflict resolution - Cross-team collaboration and code review practices

All changes were committed to the pravesh-updates branch with professional, descriptive commit messages.